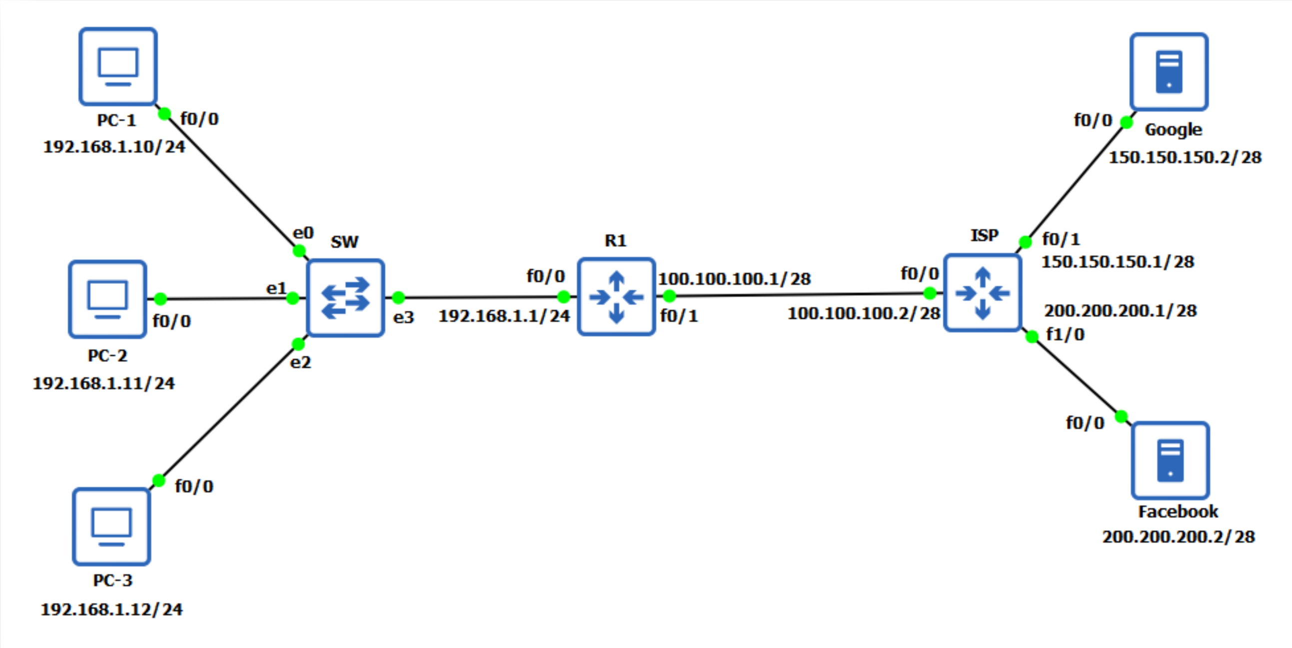
Static NAT



Task

1. Configure routers R1 & ISP with IP address as shown in topology and configure enable password as **ccna**.
2. Configure routers as Host (PC)
3. Configure routers as Host (Servers) and enable HTTP service on it.
4. Configure default routing on R1.
5. Configure static NAT to provide internet access from PC-1

Task-1: Basic IP configuration on R1 & ISP router.

R1 Configuration:

```
R1#config t
R1(config)#no ip domain-lookup
R1(config)#enable password ccna
R1(config)#int fa0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#description Link to Secure-LAN
R1(config-if)#exit
R1(config)#int fa0/1
R1(config-if)#ip address 100.100.100.1 255.255.255.240
R1(config-if)#no shut
R1(config-if)#description Link to ISP
R1(config-if)#exit
R1(config)#line vty 0 15
R1(config-line)#password cisco
```

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R1(config-line)#login
R1(config-line)#logging synchronous
R1(config-line)#exec-timeout 1 0
R1(config-line)#exit
R1(config)#exit
R1#

ISP Configuration:

ISP#config t
ISP(config)#no ip domain-lookup
ISP(config)#enable password ccna
ISP(config)#int fa0/0
ISP(config-if)#ip address 100.100.100.2 255.255.255.240
ISP(config-if)#no shut
ISP(config-if)#description Link to R1 on fa0/1
ISP(config-if)#exit
ISP(config)#int fa0/1
ISP(config-if)#ip address 150.150.150.1 255.255.255.240
ISP(config-if)#no shut
ISP(config-if)#description Link to Google Server
ISP(config-if)#exit
ISP(config)#int fa1/0
ISP(config-if)#ip address 200.200.200.1 255.255.255.240
ISP(config-if)#no shut
ISP(config-if)#description Link to Facebook Server
ISP(config-if)#exit
ISP(config)#line vty 0 15
ISP(config-line)#password cisco
ISP(config-line)#login
ISP(config-line)#logging synchronous
ISP(config-line)#exec-timeout 1 0
ISP(config-line)#exit
ISP(config)#exit
ISP#

Task-2: Configure routers as Host (PC)

PC-1 Configuration:

PC-1#config t
PC-1(config)#no ip domain-lookup
PC-1(config)#no ip routing
PC-1(config)#ip default-gateway 192.168.1.1
PC-1(config)#int fa0/0
PC-1(config-if)#ip address 192.168.1.10 255.255.255.0
PC-1(config-if)#no shut
PC-1(config-if)#description PC-1
PC-1(config-if)#exit
PC-1(config)#exit
PC-1#

PC-2 Configuration:

PC-2#config t
PC-2(config)#no ip domain-lookup
PC-2(config)#no ip routing
PC-2(config)#ip default-gateway 192.168.1.1
PC-2(config)#int fa0/0
PC-2(config-if)#ip address 192.168.1.11 255.255.255.0
PC-2(config-if)#no shut
PC-2(config-if)#description PC-2
PC-2(config-if)#exit
PC-2(config)#exit
PC-2#

PC-3 Configuration:

PC-3#config t
PC-3(config)#no ip domain-lookup
PC-3(config)#no ip routing
PC-3(config)#ip default-gateway 192.168.1.1
PC-3(config)#int fa0/0
PC-3(config-if)#ip address 192.168.1.12 255.255.255.0
PC-3(config-if)#no shut
PC-3(config-if)#description PC-3
PC-3(config-if)#exit
PC-3(config)#exit
PC-3#

Task-3: Configure routers as Host (Servers) and enable HTTP service on it.

Google Server Configuration:

Google#config t
Google(config)#no ip domain-lookup
Google(config)#no ip routing
Google(config)#enable password ccna
Google(config)#username user1 secret pass1
Google(config)#ip http server
Google(config)#ip http secure-client-auth
Google(config)#ip default-gateway 150.150.150.1
Google(config)#int fa0/0
Google(config-if)#ip address 150.150.150.2 255.255.255.240
Google(config-if)#no shut
Google(config-if)#description Google Server
Google(config-if)#exit
Google(config)#line vty 0 15
Google(config-line)#login local
Google(config-line)#transport input ssh telnet
Google(config-line)#exit
Google(config)#exit
Google#

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Facebook Server Configuration:

Facebook#config t
Facebook(config)#no ip domain-lookup
Facebook(config)#no ip routing
Facebook(config)#enable password ccna
Facebook(config)#username user1 secret pass1
Facebook(config)#ip http server
Facebook(config)#ip http secure-client-auth
Facebook(config)#ip default-gateway 200.200.200.1
Facebook(config)#int fa0/0
Facebook(config-if)#ip address 200.200.200.2 255.255.255.240
Facebook(config-if)#no shut
Facebook(config-if)#description Google Server
Facebook(config-if)#exit
Facebook(config)#line vty 0 15
Facebook(config-line)#login local
Facebook(config-line)#transport input ssh telnet
Facebook(config-line)#exit
Facebook(config)#exit
Facebook#

Task-4: Configure default routing on R1.

R1#config t
R1(config)#ip route 0.0.0.0 0.0.0.0 100.100.100.2
R1(config)#exit
R1#

✓ Verification & Testing:

R1#sh ip int brief | exclude unassign

```
Interface          IP-Address      OK? Method Status        Protocol
FastEthernet0/0    192.168.1.1     YES manual  up             up
FastEthernet0/1    100.100.100.1   YES manual  up             up
R1#
```

ISP#sh ip int brief | exclude unassign

```
Interface          IP-Address      OK? Method Status        Protocol
FastEthernet0/0    100.100.100.2   YES manual  up             up
FastEthernet0/1    150.150.150.1   YES manual  up             up
FastEthernet1/0    200.200.200.1   YES manual  up             up
ISP#
```

R1#sh ip route static

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
```

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
+ - replicated route, % - next hop override

Gateway of last resort is 100.100.100.2 to network 0.0.0.0

S* 0.0.0.0/0 [1/0] via 100.100.100.2
R1#

Ping Google and Facebook server from PC-1, PC-2 and PC-3

PC-1#ping 150.150.150.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 150.150.150.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
PC-1#

PC-1#ping 200.200.200.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 200.200.200.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
PC-1#

PC-2#ping 150.150.150.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 150.150.150.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
PC-2#

PC-2#ping 200.200.200.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 200.200.200.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
PC-2#

PC-3#ping 150.150.150.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 150.150.150.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
PC-3#

PC-3#ping 200.200.200.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 200.200.200.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
PC-3#

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Task-5: Configure static NAT to provide internet access from PC-1

```
R1#config t
R1(config)#ip nat inside source static 192.168.1.10 100.100.100.3
R1(config)#int fa0/0
R1(config-if)#ip nat inside
R1(config-if)#exit
R1(config)#int fa0/1
R1(config-if)#ip nat outside
R1(config-if)#exit
R1(config)#exit
R1#
```

- ✓ **Verification & Testing:**

R1#sh ip nat translations

```
Pro Inside global     Inside local     Outside local     Outside global
--- 100.100.100.3     192.168.1.10     ---               ---
```

Start NAT debugging on R1 and ping Google and Facebook server from PC-1, PC-2 and PC-3

R1#debug ip nat

```
IP NAT debugging is on
R1#
```

PC-1#ping 150.150.150.2

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 150.150.150.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/59/72 ms
PC-1#
```

PC-1#ping 200.200.200.2

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 200.200.200.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/56/64 ms
PC-1#
```

PC-1#telnet 150.150.150.2 80

```
Trying 150.150.150.2, 80 ... Open

PC-1#
```

PC-1#telnet 200.200.200.2 80

```
Trying 200.200.200.2, 80 ... Open

PC-1#
```

PC-2#ping 150.150.150.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 150.150.150.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
PC-2#

PC-2#ping 200.200.200.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 200.200.200.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
PC-2#

PC-3#ping 150.150.150.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 150.150.150.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
PC-3#

PC-3#ping 200.200.200.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 200.200.200.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
PC-3#

R1#sh ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
icmp	100.100.100.3:3	192.168.1.10:3	150.150.150.2:3	150.150.150.2:3
icmp	100.100.100.3:4	192.168.1.10:4	200.200.200.2:4	200.200.200.2:4
tcp	100.100.100.3:18344	192.168.1.10:18344	200.200.200.2:80	200.200.200.2:80
tcp	100.100.100.3:21249	192.168.1.10:21249	200.200.200.2:80	200.200.200.2:80
tcp	100.100.100.3:24751	192.168.1.10:24751	200.200.200.2:80	200.200.200.2:80
tcp	100.100.100.3:51678	192.168.1.10:51678	150.150.150.2:80	150.150.150.2:80
tcp	100.100.100.3:59216	192.168.1.10:59216	150.150.150.2:80	150.150.150.2:80
tcp	100.100.100.3:60594	192.168.1.10:60594	150.150.150.2:80	150.150.150.2:80
---	100.100.100.3	192.168.1.10	---	---

R1#

R1#show ip nat statistics
Total active translations: 3 (1 static, 2 dynamic; 2 extended)
Peak translations: 5, occurred 00:06:45 ago
Outside interfaces:
  FastEthernet0/1
Inside interfaces:
  FastEthernet0/0
Hits: 78 Misses: 0
CEF Translated packets: 78, CEF Punted packets: 0
Expired translations: 4

```
Dynamic mappings:
Total doors: 0
Appl doors: 0
Normal doors: 0
Queued Packets: 0

R1#
*Nov 16 20:52:52.299: NAT*: s=192.168.1.10->100.100.100.3, d=150.150.150.2 [35]
*Nov 16 20:52:52.339: NAT*: s=150.150.150.2, d=100.100.100.3->192.168.1.10 [35]
*Nov 16 20:52:52.351: NAT*: s=192.168.1.10->100.100.100.3, d=150.150.150.2 [36]
*Nov 16 20:52:52.383: NAT*: s=150.150.150.2, d=100.100.100.3->192.168.1.10 [36]
*Nov 16 20:52:52.411: NAT*: s=192.168.1.10->100.100.100.3, d=150.150.150.2 [37]
*Nov 16 20:52:52.443: NAT*: s=150.150.150.2, d=100.100.100.3->192.168.1.10 [37]
*Nov 16 20:52:52.475: NAT*: s=192.168.1.10->100.100.100.3, d=150.150.150.2 [38]
*Nov 16 20:52:52.523: NAT*: s=150.150.150.2, d=100.100.100.3->192.168.1.10 [38]
*Nov 16 20:52:52.539: NAT*: s=192.168.1.10->100.100.100.3, d=150.150.150.2 [39]
*Nov 16 20:52:52.583: NAT*: s=150.150.150.2, d=100.100.100.3->192.168.1.10 [39]

R1#
*Nov 16 20:53:04.431: NAT*: s=192.168.1.10->100.100.100.3, d=200.200.200.2 [40]
*Nov 16 20:53:04.471: NAT*: s=200.200.200.2, d=100.100.100.3->192.168.1.10 [40]
*Nov 16 20:53:04.511: NAT*: s=192.168.1.10->100.100.100.3, d=200.200.200.2 [41]
*Nov 16 20:53:04.547: NAT*: s=200.200.200.2, d=100.100.100.3->192.168.1.10 [41]
*Nov 16 20:53:04.579: NAT*: s=192.168.1.10->100.100.100.3, d=200.200.200.2 [42]
*Nov 16 20:53:04.599: NAT*: s=200.200.200.2, d=100.100.100.3->192.168.1.10 [42]
*Nov 16 20:53:04.643: NAT*: s=192.168.1.10->100.100.100.3, d=200.200.200.2 [43]
*Nov 16 20:53:04.671: NAT*: s=200.200.200.2, d=100.100.100.3->192.168.1.10 [43]
*Nov 16 20:53:04.703: NAT*: s=192.168.1.10->100.100.100.3, d=200.200.200.2 [44]
*Nov 16 20:53:04.727: NAT*: s=200.200.200.2, d=100.100.100.3->192.168.1.10 [44]

R1#
*Nov 16 20:54:53.675: NAT*: s=192.168.1.10->100.100.100.3, d=150.150.150.2 [50551]
*Nov 16 20:54:53.719: NAT*: s=150.150.150.2, d=100.100.100.3->192.168.1.10 [34784]
*Nov 16 20:54:53.731: NAT*: s=192.168.1.10->100.100.100.3, d=150.150.150.2 [50552]
*Nov 16 20:54:53.735: NAT*: s=192.168.1.10->100.100.100.3, d=150.150.150.2 [50553]
*Nov 16 20:54:53.967: NAT*: s=150.150.150.2, d=100.100.100.3->192.168.1.10 [34785]
*Nov 16 20:54:54.075: NAT*: s=150.150.150.2, d=100.100.100.3->192.168.1.10 [17944]
*Nov 16 20:54:54.115: NAT*: s=192.168.1.10->100.100.100.3, d=150.150.150.2 [17538]
*Nov 16 20:54:54.531: NAT*: s=200.200.200.2, d=100.100.100.3->192.168.1.10 [9146]

R1#
*Nov 16 20:54:54.547: NAT*: s=192.168.1.10->100.100.100.3, d=200.200.200.2 [52176]
*Nov 16 20:55:04.487: NAT*: s=200.200.200.2, d=100.100.100.3->192.168.1.10 [15200]
*Nov 16 20:55:04.519: NAT*: s=192.168.1.10->100.100.100.3, d=200.200.200.2 [49486]
*Nov 16 20:55:04.519: NAT*: s=192.168.1.10->100.100.100.3, d=200.200.200.2 [49487]
*Nov 16 20:55:04.755: NAT*: s=200.200.200.2, d=100.100.100.3->192.168.1.10 [15201]
*Nov 16 20:55:06.063: NAT*: s=200.200.200.2, d=100.100.100.3->192.168.1.10 [52688]
*Nov 16 20:55:06.107: NAT*: s=192.168.1.10->100.100.100.3, d=200.200.200.2 [32358]
R1#
```

Start icmp debugging on PC-1 and ping NATed IP from Google and Facebook server.

PC-1#debug ip icmp
ICMP packet debugging is on
PC-1#

Google#ping 100.100.100.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.100.100.3, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 44/52/56 ms
Google#

Facebook#ping 100.100.100.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.100.100.3, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 44/56/64 ms
Facebook#

PC-1#
*Nov 12 20:31:47.951: ICMP: echo reply sent, src 192.168.1.10, dst 150.150.150.2, topology BASE, dscp 0 topoid 0
*Nov 12 20:31:47.995: ICMP: echo reply sent, src 192.168.1.10, dst 150.150.150.2, topology BASE, dscp 0 topoid 0
*Nov 12 20:31:48.059: ICMP: echo reply sent, src 192.168.1.10, dst 150.150.150.2, topology BASE, dscp 0 topoid 0
*Nov 12 20:31:48.123: ICMP: echo reply sent, src 192.168.1.10, dst 150.150.150.2, topology BASE, dscp 0 topoid 0
*Nov 12 20:31:48.187: ICMP: echo reply sent, src 192.168.1.10, dst 150.150.150.2, topology BASE, dscp 0 topoid 0

PC-1#
*Nov 12 20:31:52.139: ICMP: echo reply sent, src 192.168.1.10, dst 200.200.200.2, topology BASE, dscp 0 topoid 0
*Nov 12 20:31:52.195: ICMP: echo reply sent, src 192.168.1.10, dst 200.200.200.2, topology BASE, dscp 0 topoid 0
*Nov 12 20:31:52.255: ICMP: echo reply sent, src 192.168.1.10, dst 200.200.200.2, topology BASE, dscp 0 topoid 0
*Nov 12 20:31:52.315: ICMP: echo reply sent, src 192.168.1.10, dst 200.200.200.2, topology BASE, dscp 0 topoid 0
*Nov 12 20:31:52.379: ICMP: echo reply sent, src 192.168.1.10, dst 200.200.200.2, topology BASE, dscp 0 topoid 0
PC-1#

R1#sh int fa0/1 | section address
  Hardware is i82543 (Livengood), address is ca07.27a0.0006 (bia ca07.27a0.0006)
  Internet address is 100.100.100.1/28
R1#

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```
R1#sh arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  100.100.100.1           -   ca07.27a0.0006  ARPA   FastEthernet0/1
Internet  100.100.100.2           8   ca04.2360.0008  ARPA   FastEthernet0/1
Internet  100.100.100.3           -   ca07.27a0.0006  ARPA   FastEthernet0/1
Internet  192.168.1.1             -   ca07.27a0.0008  ARPA   FastEthernet0/0
Internet  192.168.1.10           16   ca01.1f60.0008  ARPA   FastEthernet0/0
Internet  192.168.1.11           16   ca02.1be8.0008  ARPA   FastEthernet0/0
Internet  192.168.1.12           16   ca03.287c.0008  ARPA   FastEthernet0/0
R1#
```

To Remove Static NAT

```
R1#config t
R1(config)#no ip nat inside source static 192.168.1.10 100.100.100.3
Static entry in use, do you want to delete child entries? [no]: yes
R1(config)#int fa0/0
R1(config-if)#no ip nat inside
R1(config-if)#exit
R1(config)#int fa0/1
R1(config-if)#no ip nat outside
R1(config-if)#exit
R1(config)#exit
R1#
```

Important Commands:

Sh ip int brief | exclude unassign
Sh ip route static
sh ip nat translations
sh ip nat statistics
sh arp
debug ip icmp
debug ip nat

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